

Compact SHPB System for Intermediate and High Strain Rate Plasticity and Fracture Testing of Sheet Metal

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Abstract A new set-up is proposed to perform high strain rate tension experiments on sheet metal using a compression Hopkinson bar system on the input side. With the help of a custom-made load inversion device, the compression loading pulse is converted into tensile loading of the specimen boundary. A tensile output bar is used to measure the tensile force acting on the specimen boundaries. A high speed camera system is employed to measure the displacement history at the specimen level through planar digital image correlation. The output bar is positioned on top of the input bar. As a result, the valid experiment duration of the proposed system is twice as long as that of conventional Kolsky systems. It therefore facilitates the execution of intermediate strain rate ($\sim 100/\text{s}$) experiments without increasing total system length. Numerical simulations are carried out to assess the effect of spurious bending effects that are introduced through the eccentricity of the input and output bar axes. In addition, experiments are performed on straight and notched specimens to demonstrate the characterization of the rate dependent plasticity and fracture properties of a 1.06 mm thick DP780 steel.

Keywords Split Hopkinson pressure bar · High strain rate · Tension experiments · Sheet metal · Ductile fracture

Introduction

High strain rate tension experiments are either performed with tensile Hopkinson bars systems [1–12] or with compression Hopkinson bar systems provided that the incident loading pulse is converted from compression to tension before it reaches the specimen gage section ([13–21]). Dunand et al. [22] have recently presented a Load Inversion Device (LID) which allows for nearly oscillation-free force-history measurements during high strain rate tensile experiments with Split Hopkinson Pressure Bar (SHPB) systems (Fig. 1(a)). One key feature of their system is the use of two parallel output bars to measure the force history. Furthermore, their system includes double inversion, i.e. firstly compression-to-tension inversion from the input bar to the specimen, and secondly tension-to-compression inversion from the specimen to the output bar.

This technical note presents an extension of the work done by Dunand et al. [22] and describes a new enhanced LID which is used in conjunction with a compressive input bar and a tensile output bar (Fig. 1(b) and (c)). In other words, only one load inversion is required (compression-to-tension inversion from the input bar to the specimen). As compared to the test set-up proposed by Dunand et al. [22], the new system has the advantage of using only one output bar which is positioned on top of the input bar. The resulting system is therefore not only much easier to maneuver, but its total length (sum of striker, input bar, output bar lengths) is also 50 % shorter than that of a conventional Kolsky [23] bar system for the same valid experiment duration. Conversely, it allows for longer testing durations for the same total system length, i.e. it facilitates the realization of intermediate strain rate fracture experiments. As a sequel, we describe the experimental set-up and procedures, before presenting selected numerical and

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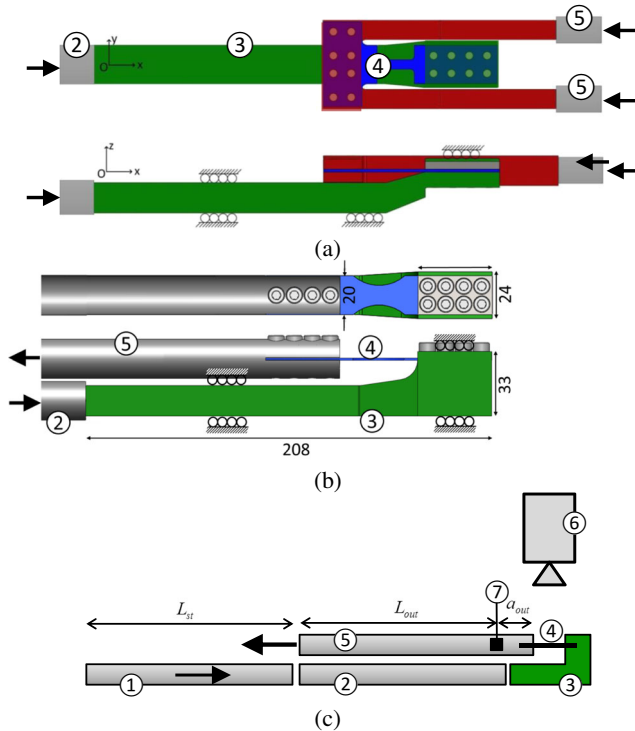


Fig. 1 Load inversion devices for use in conjunction with (a) two compressive output bars, and (b) one tensile output bar; (c) configuration of proposed SHPB system for tensile testing; The encircled numbers indicate: ① striker bar, ② input bar, ③ pusher of the LID, ④ specimen, and ⑤ output bar(s), ⑥ high speed camera and ⑦ position of strain gauge. Dimensions given are in mm

experimental results to validate the proposed high strain rate tensile testing technique for sheet metal.

Experimental Technique

Main principle

The main principle of the proposed new experimental system is shown in Fig. 1(c). A compressive loading pulse is applied to the leftmost end of a Load Inversion Device (LID, part ③ in Fig. 1(c)) using a conventional striker-input bar system. The LID then applies a tensile incident loading pulse to the specimen (part ④) which is connected to an output bar (part ⑤) for the measurement of the transmitted (tensile) wave. The main measurements throughout each experiment are:

- The history of the relative axial displacement of two points on the specimen surface (see Fig. 2(a)); it will be measured through planar digital correlation of images acquired through high speed photography (camera ⑥ in Fig. 1(c)).
- The history of the force acting on the specimen; it will be computed based on the recordings of a strain gauge (label ⑦ in Fig. 1(c)) positioned on the output bar.

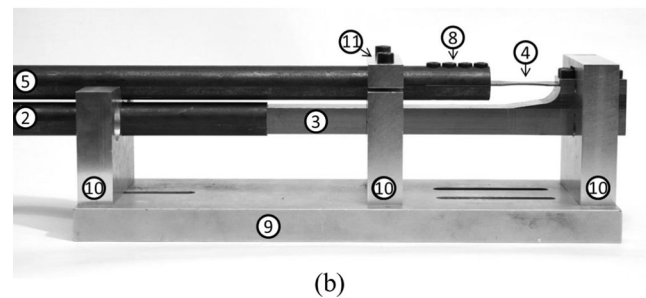
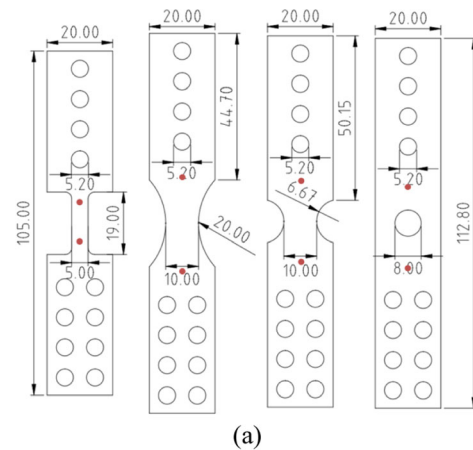


Fig. 2 (a) Specimens for experiments at high strain rates. From left to right: specimens for uniaxial tension (UT), tension with 20 mm notch radius (NT20), tension with 6 mm notch radius (NT6), and tension with a central hole (CH); the red solid dots highlight the corresponding points for displacement measurements. (b) Photograph of the Load Inversion Device: ② input Bar, ③ pusher, ④ specimen, ⑤ output bar, ⑧ grip section with slit, ⑨ base plate, ⑩ bearings, ⑪ removable clamp (useful during installation)

Design considerations

Several points are worth considering in the design of the testing system:

- *Valid duration of experiments.* The single-point force measurements become invalid with the arrival of a rightward traveling reflected wave at the position of the output bar strain gauge. The valid measurement duration T is therefore defined by the output bar configuration,

$$T = \frac{2(L_{out} - a_{out})}{c_{out}}, \quad (1)$$

with L_{out} denoting the output bar length, a_{out} denoting the distance of the strain gauge from the output bar/specimen interface (Fig. 1(c)), and c denoting the elastic wave propagation velocity. The corresponding striker bar length L_{st} would then be

$$L_{st} = Tc_{st}/2 = L_{out} - a_{out}. \quad (2)$$

with c_{st} denoting the striker bar wave speed.

- *Striker/output bar eccentricity.* The striker/output bar eccentricity is responsible for the generation of bending waves in all components of the testing system. In theory, it is equal to the sum of the input and output bar radii,

$$e = R_{in} + R_{out}. \quad (3)$$

Small input and output bar diameters are thus desirable in view of minimizing the striker/output bar eccentricity.

- It is worth noting that the input bar could be omitted entirely. It just serves as a means to transmit the loading pulse from the striker bar to the specimen, but it is neither required for speed nor for force measurements. Omitting the input bar would even reduce the total system length by almost another 50 %. Nonetheless the experiments are not performed in the “direct impact” configuration as it would result in a larger eccentricity between the striker bar and the output bar axes. Note that R_{in} would need to be replaced in equation (3) by the outer radius of the striker bar guidance system, thereby unnecessarily increasing the eccentricity.
- *Specimen clamping.* The output bar diameter also needs to be at least as wide as the specimen shoulders to allow for the successful clamping of the specimen. Furthermore, when using a slit clamping mechanism to attach the specimen to the output bar, the required thread length for the clamping screws also sets a lower limit for the output bar diameter.
- *Constant structural impedance.* Impedance variations on the input side (pusher and input bar) will not affect the measurement accuracy, but they are nonetheless undesired because of the generation of oscillations in the loading speed of the specimen shoulder.

Specific system configuration

The proposed experimental technique is applied to set up Hopkinson bar tensile experiments for characterizing the rate-dependent plasticity and fracture response of advanced high strength steel sheets.

Material and specimens

The same specimen gage section geometries as those proposed and validated by Dunand et al. [22] are used (Fig. 2a). This includes standard uniaxial tension specimens as well as specimens with cut-outs for multi-axial fracture testing. The relevant specimen characteristics for the design of the testing system are

- the specimen shoulder width of 20mm,
- the specimen thickness of 1.06mm
- maximum force anticipated prior to fracture of 15kN.

Load inversion device

Figure 2(a) shows a photograph along with a schematic of the proposed Load Inversion Device (LID). The component labeled ② is the rightmost end of the input bar. The input bar introduces a compressive pulse into the “pusher” (part ③) which in turn applies tensile loading to the attached sheet specimen (part ④). The opposite shoulder of the specimen fits into a receiving slit in the output bar (part ⑧), which is closed by tightening a set of four M5-12.9 screws. All screw heads are counter-sunk to maintain a symmetric mass distribution with respect to the bar’s axis, thereby minimizing the generation of bending waves through the specimen clamping. A screw connection with a floating clamping plate is used to establish a slip-free mechanical joint between the input bar and the specimen. The LID also features a base plate (part ⑨) with guide blocks (parts ⑩) that prevent the out-of-plane motion (bending deformation) of the pusher.

The LID pusher is made from tool steel whereas the guide blocks are made of aluminum. Assuming a friction coefficient of 0.12 between the specimen shoulder surfaces and the output bar, a clamping force of $15kN/2/0.12=66.67kN$ is required for a slip free joint. Four M5-12.9 hex cap screws are chosen, with each screw applying a theoretical maximum force of 15.6kN. For a hypothetical output bar yield strength of 300Mpa, a diameter of 8mm would be the best choice in view of minimizing the noise-to-signal ratio for force measurement. However, in order to provide sufficient thread length for the M5 screws (label ⑤) and the specimen shoulder width of 20mm, an output bar diameter of 20mm is chosen. The same argument applies when choosing the thickness of the pusher. The input bar diameter of 20mm is then chosen to approximately match the contacting cross-section of the pusher, thereby avoiding a significant impedance mismatch between the input bar and the pusher.

Hopkinson bars

The lengths of the bars are chosen such as to maximize the valid experiment duration for a given lab space of a total length of about 14m (and according to the availability of existing lab equipment):

- An output bar of length $L_{out}=4.35m$ and a measured wave speed of $c=5175m/s$ with a strain gage positioned at a distance of $a_{out}=403mm$ from the specimen interface. The system therefore provides a theoretical valid experiment duration of about 1.53ms.
- An input bar of length $L_{inp}=4.49m$ for transmitting the loading pulse from the striker to the pusher.
- A 20 mm diameter striker bar of $L_{str}=3.80m$ length; the striker bar matches the input bar impedance and generates

a 1.61ms long loading pulse (measured wave speed of $c=4725\text{m/s}$).

- For short experiments, we also make use of a $L_{str}=1.20\text{m}$ long striker bar which generates a 0.51ms long loading pulse.

Signal processing

Force measurement

The force history at the output bar/specimen interface is determined based on the recordings of the output bar strain gage. The strain at a distance of 403mm from the specimen interface is acquired at a frequency of 1MHz. Post-processing of the recorded signal is performed using the software package DAVID [24] which involves removing electrical noise (low band width filter with cut-off frequency of 100 kHz) and accounting for geometric dispersion when reconstructing the strain history $\varepsilon_{tra}[t]$ at the output bar/specimen interface. The force history then reads

$$F[t] = A_{out}E_{out}\varepsilon_{tra}[t] \quad (4)$$

with E_{out} and A_{out} denoting the output bar modulus and cross sectional area, respectively.

Displacement measurement

Prior to testing, a random speckle pattern is directly applied to the specimen surface with a speckle size of approximately 0.1 mm. A High Speed camera system (Phantom v7.3) with a 105 mm f2.8 1:1 macro lens is used to acquire a sequence of photographs for digital image correlation. A spatial resolution of 416×24 pixels turned out to be sufficient to monitor an area of 28.8×1.7 mm of the specimen surface. Images are acquired at a frequency of approximately 158 kHz (1 frame every 6.33 μs) and an exposure time of 1 μs . Two cold light sources with 150 W halogen bulbs are used to provide the required lighting. The camera is triggered by the rising edge of the electrical signal provided by a strain gage positioned at the center of the input bar.

The image correlation is performed with the commercial software VIC2D (Correlated Solutions). The data is processed using a 15×15 pixel subset and spline cubic interpolation. A virtual extensometer is used to determine the relative axial displacement $u[t]$ of two points located on the specimen surface. The red dots in Fig. 2(a) illustrate the position of the measurement points on the initial specimen geometries. The triggering input strain gage signal is recorded on the same acquisition card as the output bar strain signal. The force and displacement histories can thus be provided on the same time scale which avoids any non-physical “time shifting” when

constructing the force-displacement curve from the combination of $F[t]$ and $u[t]$,

$$F[u] = F[t] \circ u[t]. \quad (5)$$

Note that different from conventional SHB systems, the relative motion of the specimen shoulders is determined through digital image correlation instead of measuring the reflected wave. Also note that the interpretation of the latter is very difficult in the present case because of the varying pusher impedance.

Numerical Validation

The quasi-static equilibrium of the specimen gage sections in experiments with loading speeds of up to 15 m/s have been verified by Dunand et al. [22]. Here we make use of numerical simulations to validate

1. The absence of significant bending of the specimen and the output bar due to the eccentricity of the specimen axis with respect to the input bar;
2. The absence of significant oscillations in the loading velocity at the specimen boundary due to impedance variations on the input side (in particular due to the varying pusher cross-section at the point of specimen attachment)

Finite element model

The complete SHPB setup is modeled using the commercial finite element software Abaqus/Explicit [25]. Taking advantage of the symmetry of the mechanical system, only half of every component is modeled with the respective symmetry boundary condition applied. Reduced integration brick elements of the type C3D8R are used. The element sizes are:

- Input bar: $l_e=5\text{mm}$ in the axial direction and $l_h=2\text{mm}$ in the radial direction;
- Pusher and output bar: $l_e=2\text{mm}$ in the axial direction and $l_h=1\text{mm}$ in the width/radial direction;
- Specimen: $l_e=0.5\text{mm}$ in the axial and width direction, four elements in thickness direction;

The bearings are represented through rigid bodies assuming frictionless contact with the pusher. A penalty contact with a friction coefficient of 0.2 is defined between the input bar and the pusher. The specimen grip sections are connected to the pusher and the output bar with a tie constraint.

An isotropic, elasto-plastic rate-independent J_2 -plasticity model with isotropic strain hardening is employed to model the specimen material (DP780 steel); integration points are removed from the simulation when an equivalent plastic strain of 0.7 is reached. All other system components are modeled as linear elastic ($E=179\text{GPa}$, $\nu=0$, $\rho=8.03\text{g/cm}^3$ for the input bar, $E=214\text{GPa}$, $\nu=0.3$, $\rho=7.98\text{g/cm}^3$ for the output bar and $E=210\text{GPa}$, $\nu=0.3$, $\rho=7.85\text{g/cm}^3$ for the pusher). The choice of zero Poisson's ratio ($\nu=0$) avoids any geometric dispersion of the waves traveling within the input bar. Experimentally-measured velocity histories of the input-bar pusher interface are thus applied to the left-most end of the input bar. In particular, the simulations are performed for a uniaxial tension specimen for two loading scenarios:

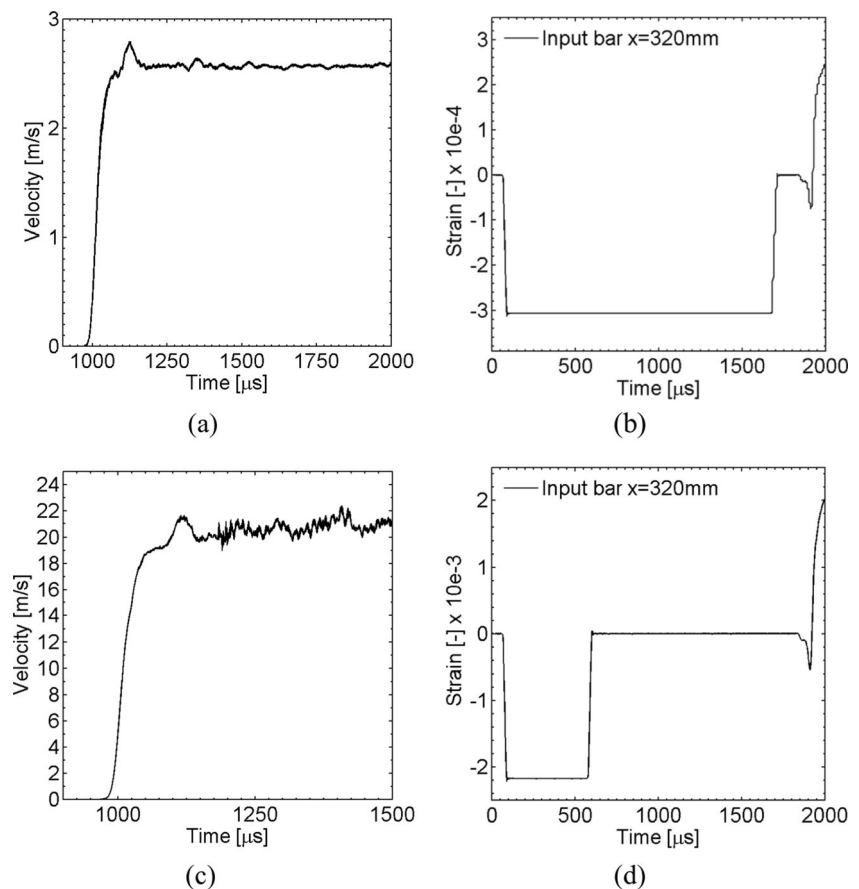
1. Slow loading: the applied velocity history (Fig. 3(a)) corresponds to that measured after the impact of a 3.8 m long striker at a speed of 2.9 m/s (Fig. 3(b));
2. Fast loading: the applied velocity history (Fig. 3(c)) corresponds to that measured after impact of a 1.2 m long striker at a speed of 20.56 m/s (Fig. 3(d));

Simulation results

Specimen loading velocity

The force levels associated with the incident waves are 126kN and 21kN for the fast and slow speed of loading, respectively. Given a maximum UT specimen resistance of 4kN, a significant portion of the incident wave signal (more than 80 %) is reflected at the specimen/pusher interface. The corresponding loading velocity of the pusher/specimen interface is therefore expected to be approximately constant which is confirmed by the computed velocity histories shown in Fig. 3(a) and (c). Both velocity-time curves show a pronounced peak lasting approximately 50 μs . In the case of the fast experiment, this peak corresponds to an increase in velocity from 20 to 22 m/s. In the case of the slow experiment, it causes an increase from 2.6 to 2.8 m/s. The peak is attributed to the initial bending of the pusher which features an impedance and axial eccentricity increase from left to right. A plot of the strain histories at the beginning of the input bar for a short striker bar (Fig. 3(b) and (d)) show that the reflected wave is of compressive nature at the beginning of the experiment. This is evidence of the increased impedance of the massive clamping end of the pusher. After the peak in the input velocity history, the loading

Fig. 3 Simulation results for slow (first row) and fast loading (second row); the pusher/specimen interface velocity histories are shown in the left column, the strain histories in the input bar (at a distance of 320 mm from the striker/input bar interface) are shown in the second column



velocity of the specimen remains more or less constant throughout the entire experiment.

Bending effects

The possible bending at the specimen level is evaluated by comparing the evolution of the equivalent plastic strain at the top and bottom specimen surfaces (Fig. 4(a) and (b)). For both the slow and fast loading scenarios, the curves lie perfectly on top of each other which is seen as a proof of absence of plastic bending of the specimen.

In addition to plastic bending, the presence of elastic bending (vibration) is verified at the position of the output bar strain gage. The extraction of the strain histories at the top and bottom of the output bar are free from bending effects for slow loading (Fig. 4(c)). However, in the case of fast loading (Fig. 4(d)), an oscillation becomes apparent in the strain signals after the axial force maximum, with a maximum difference in strain of $\Delta\varepsilon = 1.5 \times 10^{-5}$. According to elasticity theory, the measured elastic bending moment in the output bar is

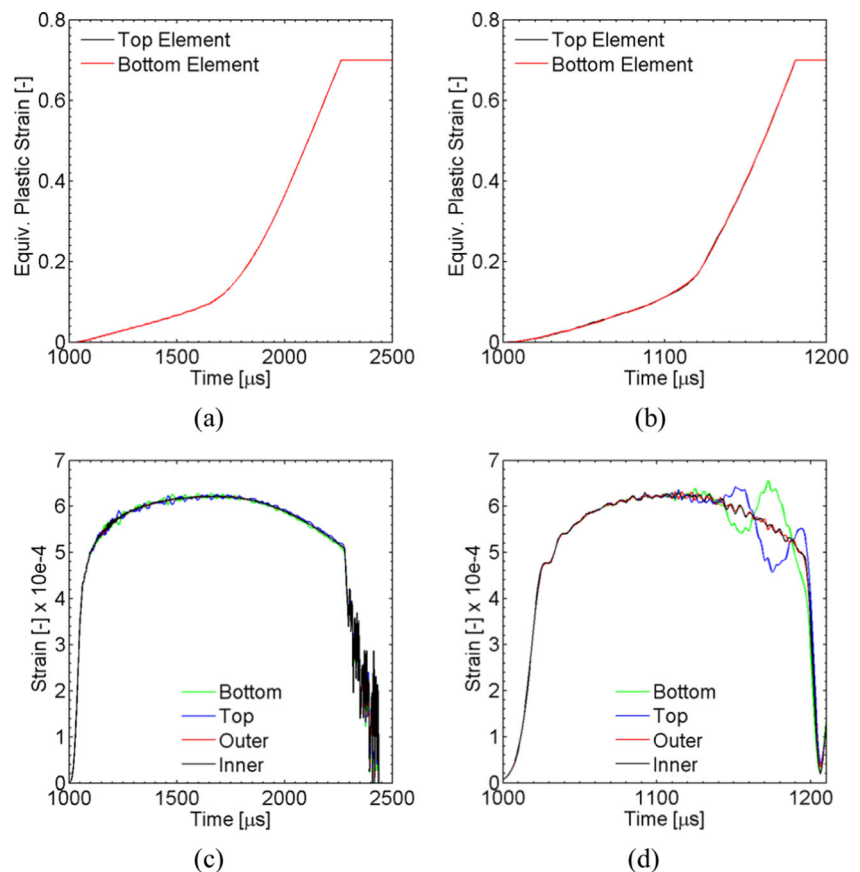
$$M = EI\kappa = E \left(\frac{\pi D^4}{64} \right) \left(\frac{\Delta\varepsilon}{D} \right) = 1.2 Nm. \quad (6)$$

Due to the absence of any eccentricity on the output side, the bending wave must have been transmitted through the specimen. The plastic bending moment of the specimen at force maximum is approximately

$$M_{pl} = \frac{bt^2}{4} \sigma_y. \quad (7)$$

with $\sigma_y \cong 850 \text{ MPa}$ denoting the yield stress at the onset of necking (Considère strain). Upon evaluation of (7) for the uniaxial tensile specimens, we find $M_{pl} \cong 1 \text{ Nm}$. The fact that the bending moment in the output bar is higher than the plastic bending moment of the specimen is explained by the presence of a shear force (in the vertical direction) of up to 60 N. Even though the effect of the shear force is expected to be negligible at the specimen level (as compared to an axial force of $>2000 \text{ N}$), the bending in the output bar needs to be taken into account when measuring the axial force. By default, the strain gages are symmetrically installed on both sides of the bar's neutral axis which removes possible bending effects from the strain history recording, i.e. the reported output bar strain histories correspond to the average axial strain and are hence suitable for calculating the axial force acting on the specimen.

Fig. 4 Analysis of bending effects associated with the input bar eccentricity: *left column* at a striker speed of 2.9 m/s and *right column* at a striker speed of 20.56 m/s; The *top row* shows the evolution of the equivalent plastic strain at the specimen top and bottom surfaces (note that the pronounced change in strain rate is due to through thickness necking), the *bottom row* shows strain histories at the position of the output bar strain gage. The histories are extracted from four elements positioned on the top ($z=D/2$), the bottom ($z=-D/2$), and the center ($z=0$, inner and outer) of the bar



Applications

Material

Experiments are carried out on a 1.06 mm thick dual phase steel DP780 provided by US Steel. All specimens are extracted with their axis aligned with the sheet rolling direction using water jet cutting. The results from static tests on a universal testing machine at a strain rate of 0.001/s are also included in most graphs for reference.

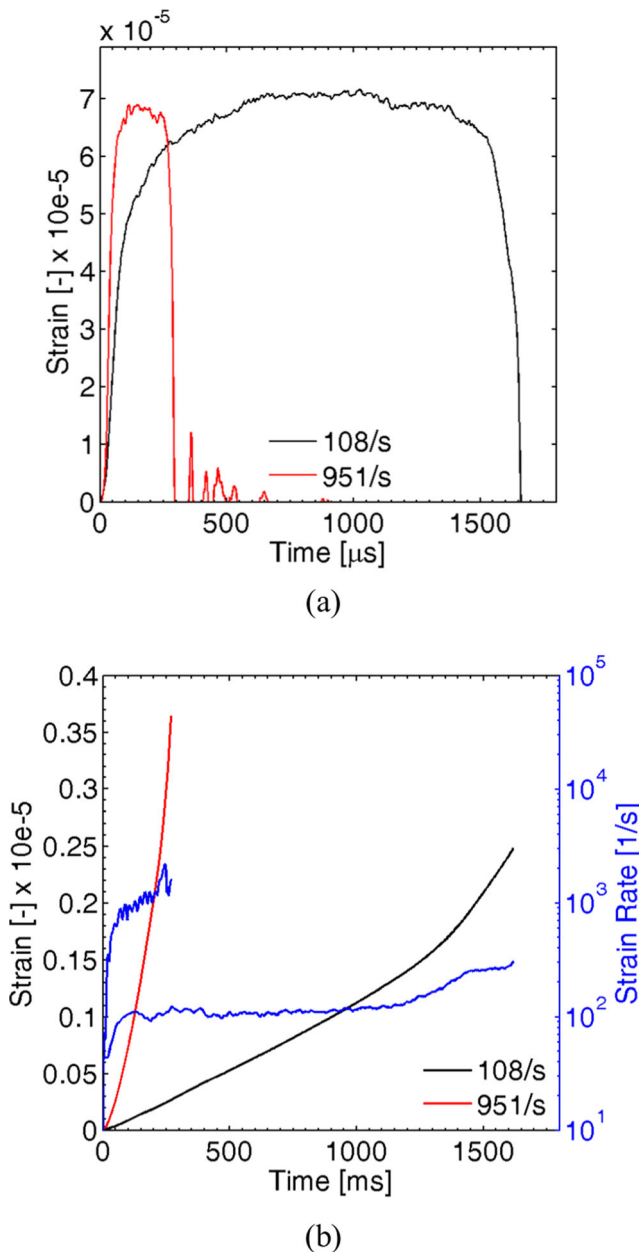


Fig. 5 Experimental results: (a) strain histories measured at the position of the output bar strain gage, (b) axial engineering strain history and strain rate history of the specimen as determined through DIC (bottom row)

Rate effect on plasticity

Uniaxial tension experiments on the UT specimen shown in Fig. 2 are carried out with the long striker (pulse duration approximately 1610 μ s) at a striker impact velocity of 3.27 m/s, and with the short striker (pulse duration approximately 551 μ s) at an impact velocity of 19.8 m/s. Figure 5(a) and (b) show the measured strain histories from the strain gage on the output bar after removal of the electrical noise. As predicted in the simulations, only minor oscillations are observed. The axial strain histories as determined through DIC at a frequency of 158 kHz are shown in Fig. 5(b). The average axial strain rates are 108/s and 951/s for the slow and fast experiments, respectively. Thanks to the long valid testing duration, the specimen could be successfully loaded all the way to fracture at both intermediate and high strain rates.

Figure 6(a) shows the measured engineering stress-strain curves for three different strain rates: 0.001/s, 108/s and 951/s.

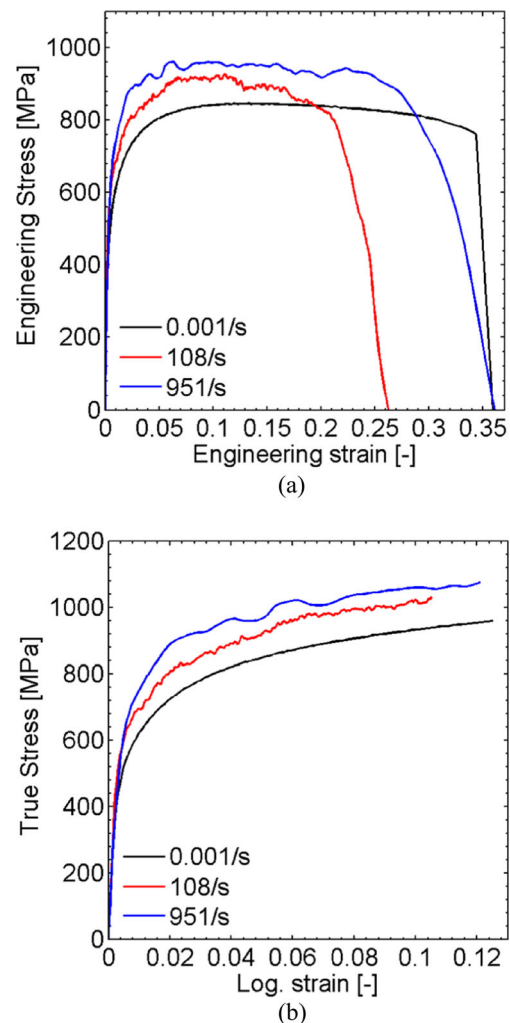


Fig. 6 Results from experiments with uniaxial tensile specimen at quasi-static strain rate (black curves), a strain rate of 108/s (red curves) and 951/s (blue curves): (a) Engineering stress-strain curves, and (b) true stress-logarithmic strain curves

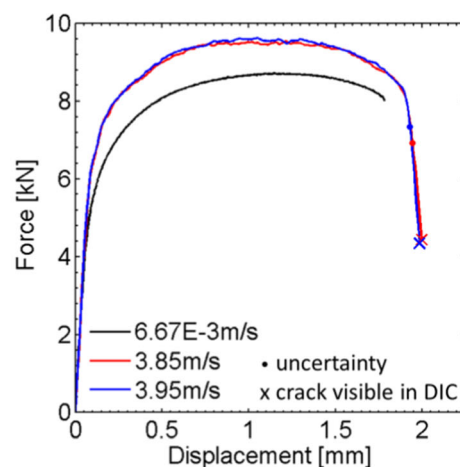
As it was already observed in the strain histories, the stress strain curve exhibits only minor oscillations of the order of ± 5 MPa, corresponding to ± 25 N in the force signal. Figure 6(b) shows the true stress- logarithmic strain curve for the three different strain rates. All curves are truncated at the point of the onset of necking (force maximum) in the experiment. The experimental data clearly reveals the effect of the strain rate on the material's stress-strain response. At the same time, the proximity of the curves for 108/s and 951/s underlines the importance of nearly oscillation-free measurements to detect and quantify the effect of strain rate on the tested material.

Rate effect on ductile fracture

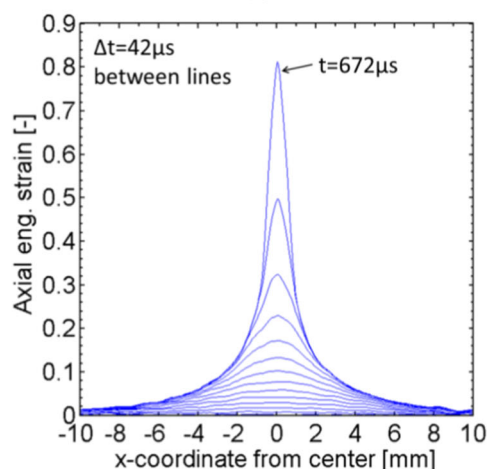
Intermediate strain rate (~ 100 /s) fracture experiments are extremely difficult to achieve since the loading rate is too fast for the use of hydraulic systems, while the testing duration is too long for conventional lab size Hopkinson bar systems. Due to the long valid duration of the experiments, the proposed experimental device is particularly suitable to perform intermediate and high strain rate fracture experiments.

Possible specimen geometries for fracture testing are shown in Fig. 2(a). For illustration purposes, we perform two high strain rate fracture experiments on notched tension specimens extracted from the DP780 steel sheets at a striker speed of approximately 6.4 m/s. The corresponding DIC measured loading velocities at the specimen level are 3.82 and 3.95 m/s. The superposition of the measured force-displacement curves (Fig. 7(a)) demonstrates the excellent repeatability of the experimental procedure. The curves are truncated at the point where the first crack was observed in the DIC measurements. For an image acquisition frequency of 120 kHz at a resolution of 512×32 pixel, the displacement between subsequent frames is 0.05 mm, which corresponds to an uncertainty of $0.05\text{ mm}/1.9\text{ mm} = 2.5\%$ in the measured displacement to fracture.

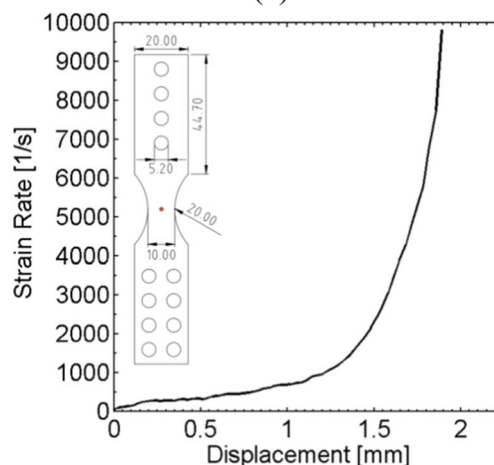
The evolution of the surface strain profile along the specimen axis is shown in Fig. 7(b). It shows intensifying localization of deformation at the specimen center. When plotting the surface strain at the specimen center as a function of the displacement, the localization of deformation becomes apparent as an increase in the strain rate from about 285/s in the beginning to over 9800/s at the end of the experiment (Fig. 7c).



(a)



(b)



(c)

Fig. 7 Results from tensile experiments with notched specimens with a radius of 20 mm. (a) Summary of force-displacement curves at slow static loading (black line), and for relative specimen boundary velocities of 3.85 m/s (red line) and 3.95 m/s (blue line). The cross symbol indicates the instant at which a first crack becomes visible in the DIC images, while the dot represents the associated uncertainty in the fracture displacement (~ 0.05 mm). (b) Evolution of the surface strain along the specimen axis (specimen tested at 3.85 m/s). (c) Strain rate as a function of the displacement in the center of the specimen as determined from DIC (specimen tested at 3.85 m/s). Note that the pronounced change in strain rate is due to through thickness necking.

Summary

A novel load inversion device has been proposed to perform dynamic tension experiments on sheet specimens using a

conventional compressive striker/input bar system in conjunction with a tensile output bar. The output bar is positioned on top of the input bar which results in a remarkably compact Hopkinson bar system configuration. The proposed experimental technique has been validated through numerical analysis of the entire testing system. The results from uniaxial tension experiments on specimens extracted from a DP780 steel are presented for intermediate (108/s) and high strain rates (951/s). In addition, the results from ductile fracture experiments on notched tensile specimens are shown.

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